

Problem 15.11

Compare the total amount of interest that would be paid on a \$3,000 loan over a 6-year period with an effective rate of interest of 7.5% per annum, under each of the following repayment plans:

- (a) The entire loan plus accumulated interest is paid in one lump sum at the end of 6 years.
- (b) Interest is paid each year as accrued, and the principal is repaid at the end of 6 years.
- (c) The loan is repaid with level payments at the end of each year over the 6-year period.

Solution

The loan amount is 3000, the annual effective interest rate is

$$i = 0.075,$$

and the term is 6 years.

Part (a)

If the entire loan plus accumulated interest is paid at the end of 6 years, the amount repaid is

$$3000(1.075)^6.$$

Thus, the total interest paid is

$$3000(1.075)^6 - 3000.$$

Therefore,

$$3000(1.075)^6 - 3000 \approx 1629.904577.$$

So the total interest paid is

$$\boxed{\$1,629.90}.$$

Part (b)

If interest is paid each year as accrued, then the annual interest payment is

$$3000(0.075) = 225.$$

There are 6 annual interest payments, so the total interest paid is

$$6(225) = 1350.$$

Hence, the total interest paid is

$$\boxed{\$1,350.00}.$$

Part (c)

Let X be the level annual payment. Since the loan is repaid with 6 level payments at the end of each year,

$$3000 = Xa_{\overline{6}|}$$

Using

$$a_{\overline{6}|} = \frac{1 - (1.075)^{-6}}{0.075},$$

we get

$$X = \frac{3000}{\frac{1 - (1.075)^{-6}}{0.075}}.$$

Thus,

$$X \approx 639.134674.$$

The total amount paid over the 6 years is

$$6X \approx 6(639.134674) = 3834.808042.$$

Therefore, the total interest paid is

$$3834.808042 - 3000 = 834.808042.$$

Hence, the total interest paid is

$$\boxed{\$834.81}.$$

Comparison

The total interest paid under each repayment plan is:

Plan	Total Interest Paid
(a)	\$1,629.90
(b)	\$1,350.00
(c)	\$834.81

Thus, among the three plans, the level-payment plan produces the smallest total amount of interest paid.